Lab Report: Text classification with neural networks

1 Introduction

In this report we compare several models on standard benchmarks. Each experiment was repeated 3 times with different random seeds. All experiments were run on a single GPU with 16 GB of memory.

2 Results

Figure 1 shows the accuracy by dataset.

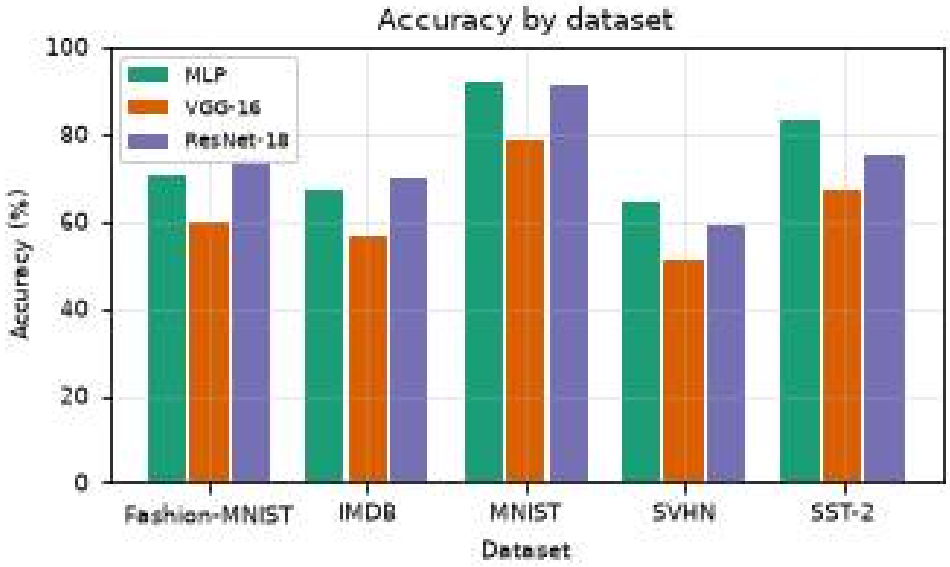


Figure 1: Accuracy by dataset for the models in this experiment.

MLP reaches 92% accuracy on MNIST. VGG-16 reaches 83.0% accuracy on SST-2. ResNet-18 reaches 59% accuracy on SVHN (Figure 1).

Figure 2 shows the test accuracy during training.

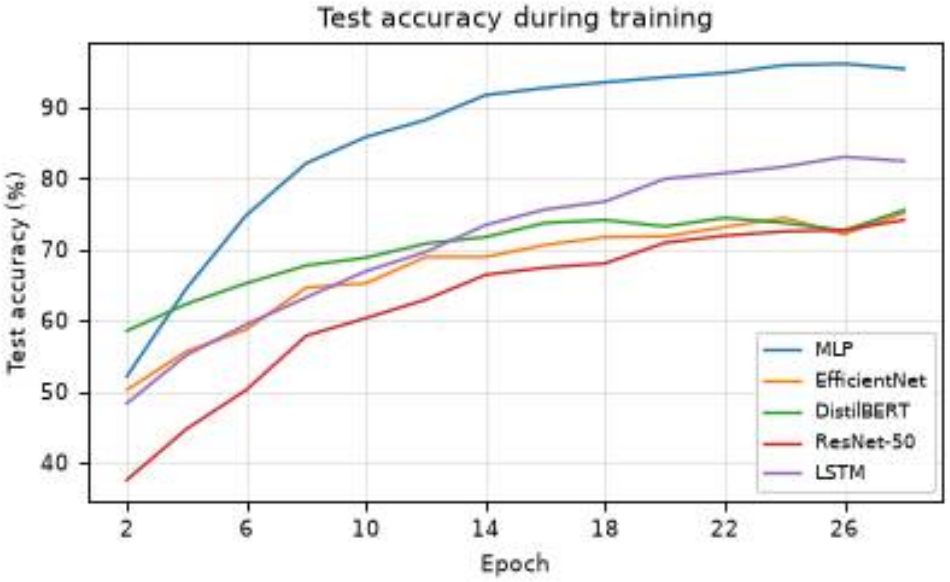


Figure 2: Test accuracy during training for the models in this experiment.

EfficientNet attains a test accuracy of 48.9% at epoch 2. LSTM outperforms ResNet-50 by 9.1 percentage points at epoch 24.

3 Discussion

Logistic Regression reaches 80.0% accuracy on SVHN. Overall, the results are in line with what we expected from the literature.